

Modeling Proportions and Compositions with Beta and Dirichlet Regression

Definitions, Interpretation, Comparison, and Empirical Application

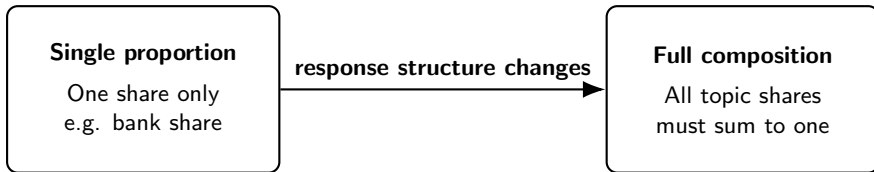
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Research Motivation

- Many applied datasets involve proportions, so ordinary Gaussian linear models are often not suitable.
- In this project, the response may be either a **single proportion** or a **full composition**.
- This leads to two related modelling choices: beta regression and Dirichlet regression.



Main question:

How should we model proportion-type data when the response changes from one share to a whole composition?

Dataset and Variables

The empirical application uses a monthly country-level topic-proportion dataset based on G7 central bank speeches. For Canada, the speech source includes *The Path to Price Stability* (Bank of Canada).

The response is a five-part topic composition: **bank**, **market**, **growth**, **govern**, and **other**.

- **bank**: central bank policy, inflation, and interest rates
- **market**: financial markets and exchange-rate conditions
- **growth**: output, demand, and labour market conditions
- **govern**: government, institutions, and public-policy issues
- **other**: residual content outside the four main topics

The main explanatory variables are:

- real GDP
- headline inflation
- USD exchange rate
- policy rate
- COVID indicator

Beta Regression for a Single Proportion

Beta regression is designed for a response variable taking values in the open interval $(0, 1)$.

Under the mean–precision parameterization,

$$Y_i \sim \text{Beta}(\mu_i \phi_i, (1 - \mu_i) \phi_i),$$

Here, μ_i is the conditional mean of the topic share, and ϕ_i is the precision parameter.

Using the logit link, the mean is related to covariates by

$$\log\left(\frac{\mu_i}{1 - \mu_i}\right) = \vec{x}_i^\top \vec{\beta},$$

where

$$\vec{x}_i = (1, \text{realGDP}_i, \text{headlineInflation}_i, \text{usdExchangeRate}_i, \text{policyRate}_i)^\top.$$

Here, $\vec{x}_i$ is the covariate vector and $\vec{\beta}$ is the coefficient vector.

A Basic Beta Regression Example

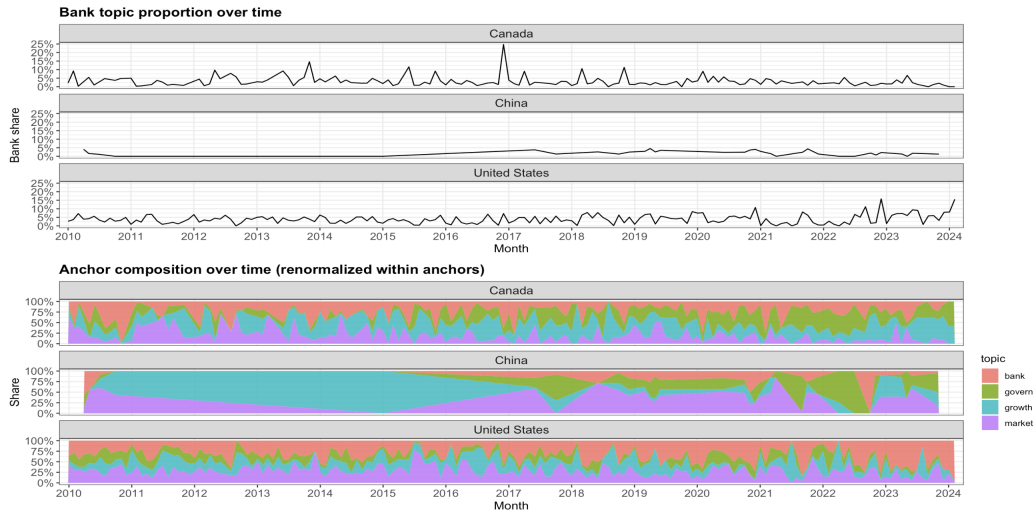
As an introductory example, I fit a beta regression model to the **bank topic share for Canada**.

Variable	Estimate	Std. Error	p-value
Intercept	-2.670	0.840	0.0015
real GDP	-0.036	0.024	0.1359
headline inflation	-0.002	0.011	0.8338
USD exchange rate	-0.209	0.611	0.7327
policy rate	-0.129	0.069	0.0607
Precision (ϕ)	50.826	6.212	< 0.001
Pseudo R^2		0.1019	
Log-likelihood		372.1	

The overall evidence is modest, which suggests that modelling one topic share alone explains only a limited part of the variation.

Why Beta Regression Is Not Enough

Beta regression is useful for one topic share only; the main question here is reallocation across topics.



Dirichlet Regression for Compositional Responses

Dirichlet regression is used when the response is the full topic composition rather than one share analysed in isolation.

Let

$$\vec{Y}_i = (Y_{i1}, \dots, Y_{iC}), \quad Y_{ic} \in (0, 1), \quad \sum_{c=1}^C Y_{ic} = 1.$$

The composition is modelled as

$$\vec{Y}_i \sim \text{Dirichlet}(\vec{\alpha}_i), \quad \alpha_{ic} = \phi_i \mu_{ic}.$$

Here, $\vec{\mu}_i$ is the expected topic composition and ϕ_i is the precision parameter.

Using the alternative baseline log-ratio parameterization,

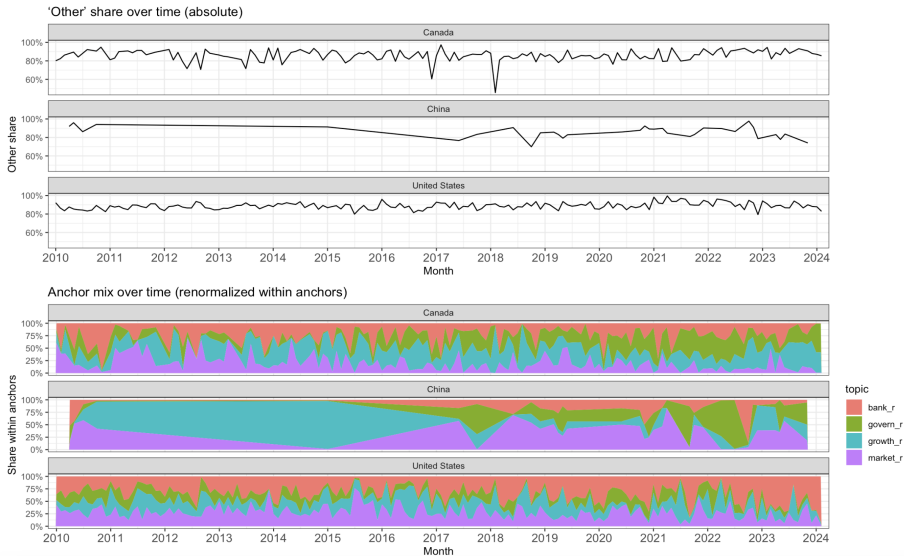
$$\log\left(\frac{\mu_{ic}}{\mu_{ib}}\right) = \vec{x}_i^\top \vec{\beta}_c, \quad c \neq b,$$

and

$$\log(\phi_i) = \vec{z}_i^\top \vec{\gamma}.$$

Here, $\vec{z}_i^\top \vec{\gamma}$ models precision, that is, how dispersion changes with covariates.

Diagnostics for the Dirichlet Specification



COVID as an Event Effect in the Linear Dirichlet Model

To justify retaining the COVID indicator, I compare two nested alternative Dirichlet regression models:

- **Model 1:** macroeconomic covariates only
- **Model 2:** macroeconomic covariates + COVID indicator

The analysis of deviance shows that adding the COVID term improves model fit substantially.

Model	Deviance	No. par.	Difference	df	p-value
Macros only	-35829	21	–	–	–
Macros + COVID	-35889	25	59.659	4	3.421×10^{-12}

So the pandemic period contributes explanatory power beyond the standard macroeconomic covariates alone.

Why Move Beyond the Linear Model?

Monthly macroeconomic effects may be nonlinear, so a purely linear model can be too restrictive.

Basis smoothing

$$x(t) = \sum_{k=1}^K c_k \phi_k(t)$$

$\phi_k(t)$ are basis functions and c_k are coefficients.

Used in the **selection stage**. Smoothness is controlled through the basis size.

Main difference: basis smoothing controls smoothness through the chosen basis, while penalized smoothing controls it directly through λ .

Penalized smoothing

$$\text{PENSSE}(c) = (\vec{y} - \Phi \vec{c})^\top W (\vec{y} - \Phi \vec{c}) + \lambda \vec{c}^\top R \vec{c}$$

The first term measures fit, and the second term penalizes roughness.

Used in the **final model**. λ controls smoothness directly.

Model Selection: Forward selection and BIC

The final model is not simply the largest candidate model. It is the version retained after the main comparison steps:

Linear baseline $\rightarrow$ Forward selection $\rightarrow$ BIC comparison $\rightarrow$ Penalized final fit

The resulting basis-selection model can be written as

$$g(\mu_i) = \alpha_{\text{country}(i)} + f_1(\text{headlineInflation}_i) + f_2(\text{policyRate}_i) + f_3(\text{usdExchangeRate}_i) + f_4(\text{realGDP}_i).$$

The BIC comparison supports using smooth terms rather than purely linear terms for the retained macroeconomic covariates:

Variable	BIC (linear)	BIC (smooth)	Δ BIC
Headline inflation	-1953563	-2015421	-61859
Policy rate	-2015421	-2046846	-31425
USD exchange rate	-2046846	-2064468	-17622
Real GDP	-2064468	-2076603	-12134

Final Penalized-Smoothed Model

Penalized smoothing was then used for the final model to control curve shape and avoid overly wiggly fits.

The final multi-country penalized-smoothing specification:

- keeps country effects,
- retains the selected smooth macroeconomic terms,
- and includes the country $\times$ policyRate interaction.

Model	AIC	BIC
Without country $\times$ policyRate	-2055859	-2054704
With country $\times$ policyRate	-2218910	-2217422
Change after adding interaction	-163051	-162718

G7 Comparison: Overall Evidence from Table 9

Table 9 shows that, for every G7 country, the single-country model has lower MAE and lower RMSE than the corresponding multi-country fit.

Country	multi MAE	multi RMSE	single MAE	single RMSE
Canada	0.0262	0.0428	0.0241	0.0402
France	0.0182	0.0278	0.0170	0.0264
Germany	0.0206	0.0300	0.0188	0.0279
Italy	0.0223	0.0341	0.0209	0.0324
Japan	0.0150	0.0222	0.0141	0.0211
UK	0.0184	0.0253	0.0177	0.0242
US	0.0164	0.0228	0.0158	0.0221

So the improvement is consistent across all seven G7 countries, which suggests that country-specific models describe national topic dynamics more accurately than a shared multi-country fit.

Japan as an Illustration of the G7 Result

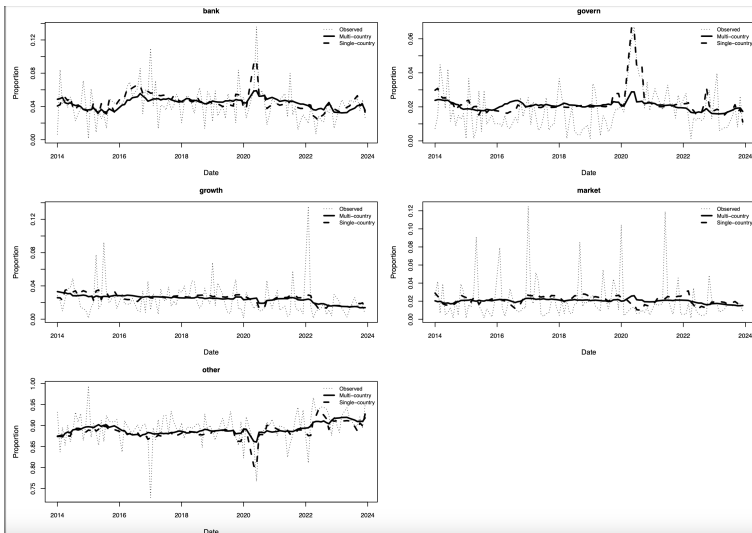


Figure 2: Japan — observed proportions and fitted values from the multi-country and single-country models.

Canada as a Second Illustration of the G7 Result

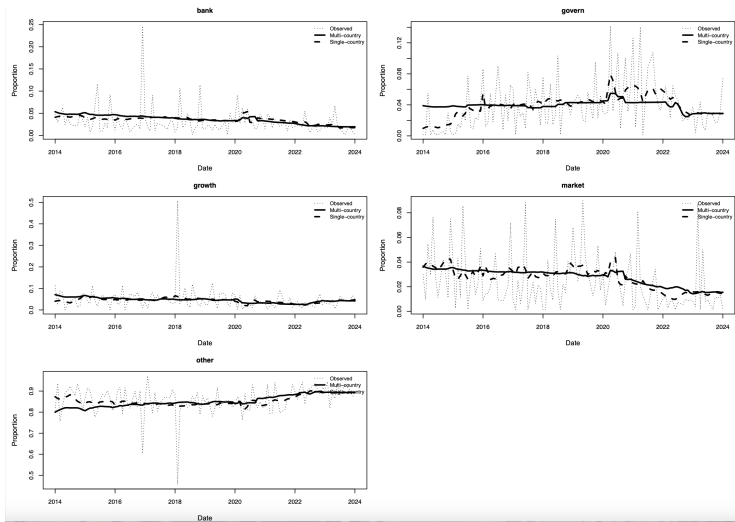


Figure 3: Canada — observed proportions and fitted values from the multi-country and single-country models.

This project compared beta regression and Dirichlet regression for modelling bounded topic proportions in a macroeconomic setting.

- Beta regression is useful when the response is a single topic share considered on its own.
- Dirichlet regression is more appropriate when the interest is in the full topic allocation, because it models the composition jointly and respects the sum-to-one constraint.
- The empirical results show that the COVID term adds explanatory power and that smoothing improves fit when linear effects are too restrictive.
- Within the G7, single-country models fit the observed topic dynamics better than the shared multi-country specification.

Overall, the project shows that the regression framework should follow the structure of the response, and that flexible nonlinear modelling can improve the analysis of macroeconomic topic allocation data.